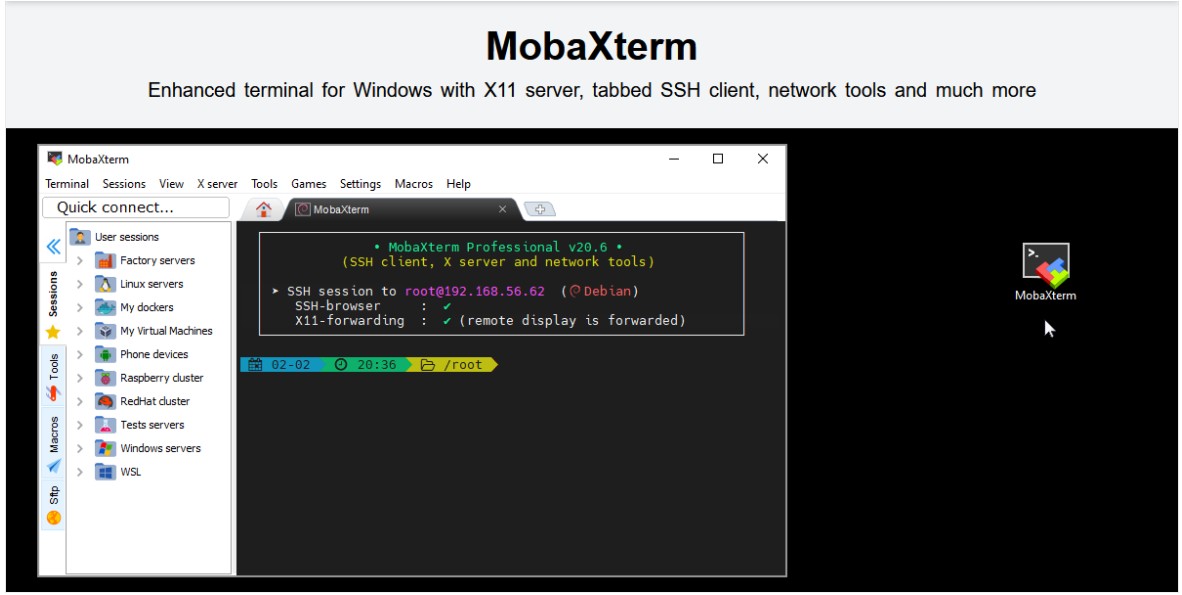
Access to the Linux server - WINDOWS

https://mobaxterm.mobatek.net/download-home-edition.html



Home Edition

Free

Full **X server** and **SSH** support

Remote desktop (RDP, VNC, Xdmcp)

Remote terminal (SSH, telnet, rlogin, Mosh)

X11-Forwarding

Automatic SFTP browser

Master password protection

Plugins support

Portable and installer versions

Full documentation

Max. **12** sessions

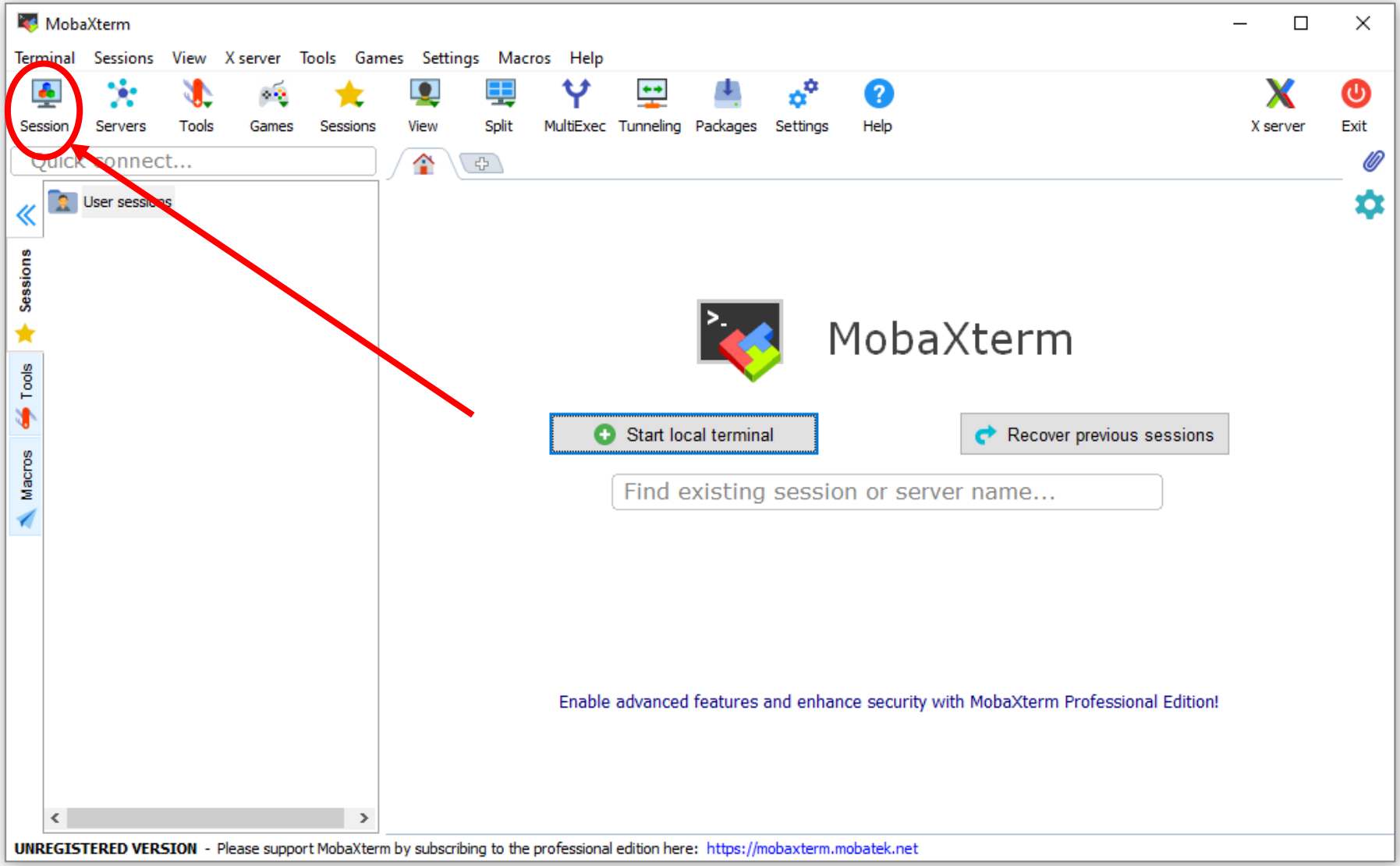
Max. **2** SSH tunnels

Max. **4** macros

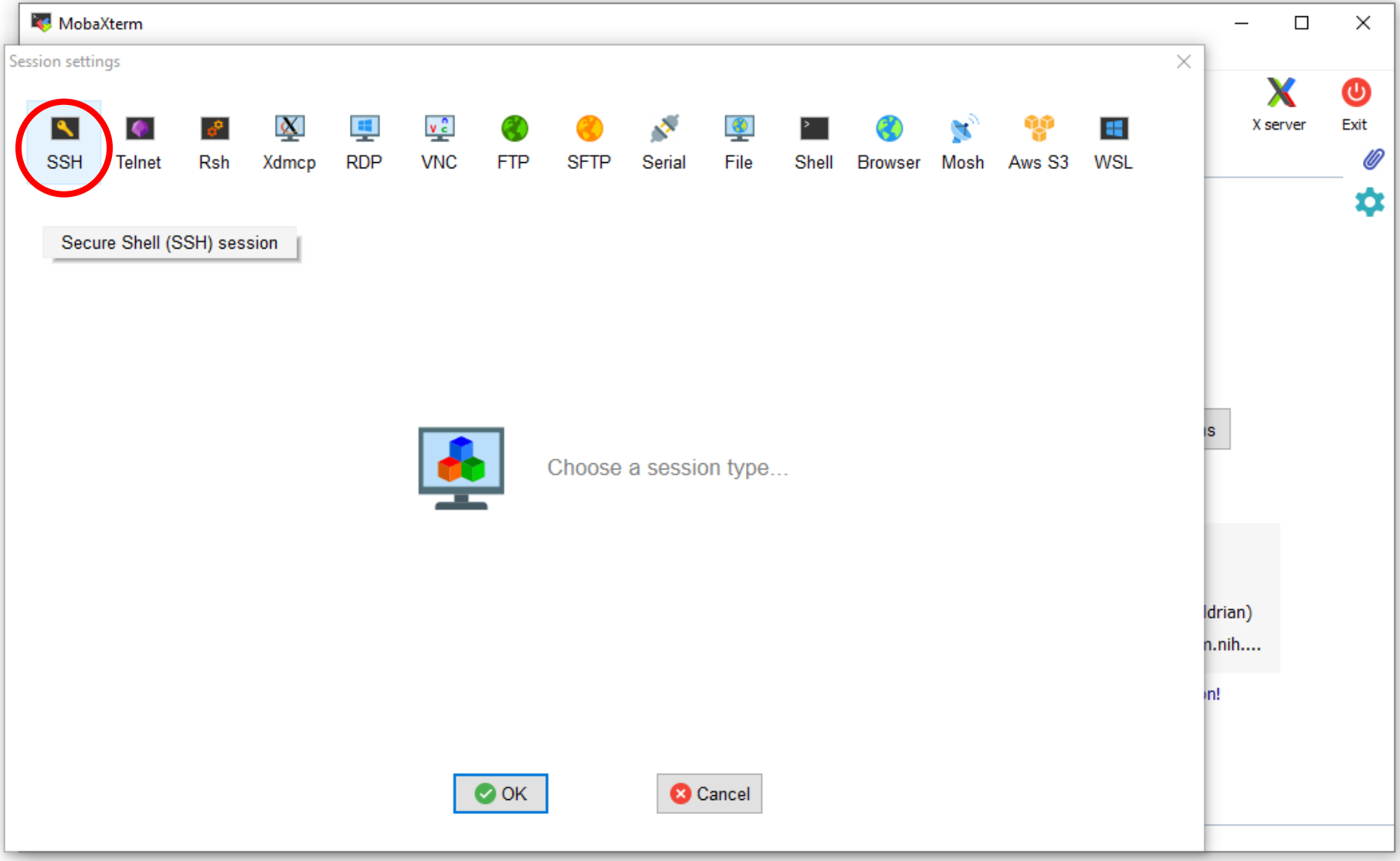
Max. **360** seconds for Tftp, Nfs and Cron

Download now

Access to the Linux server

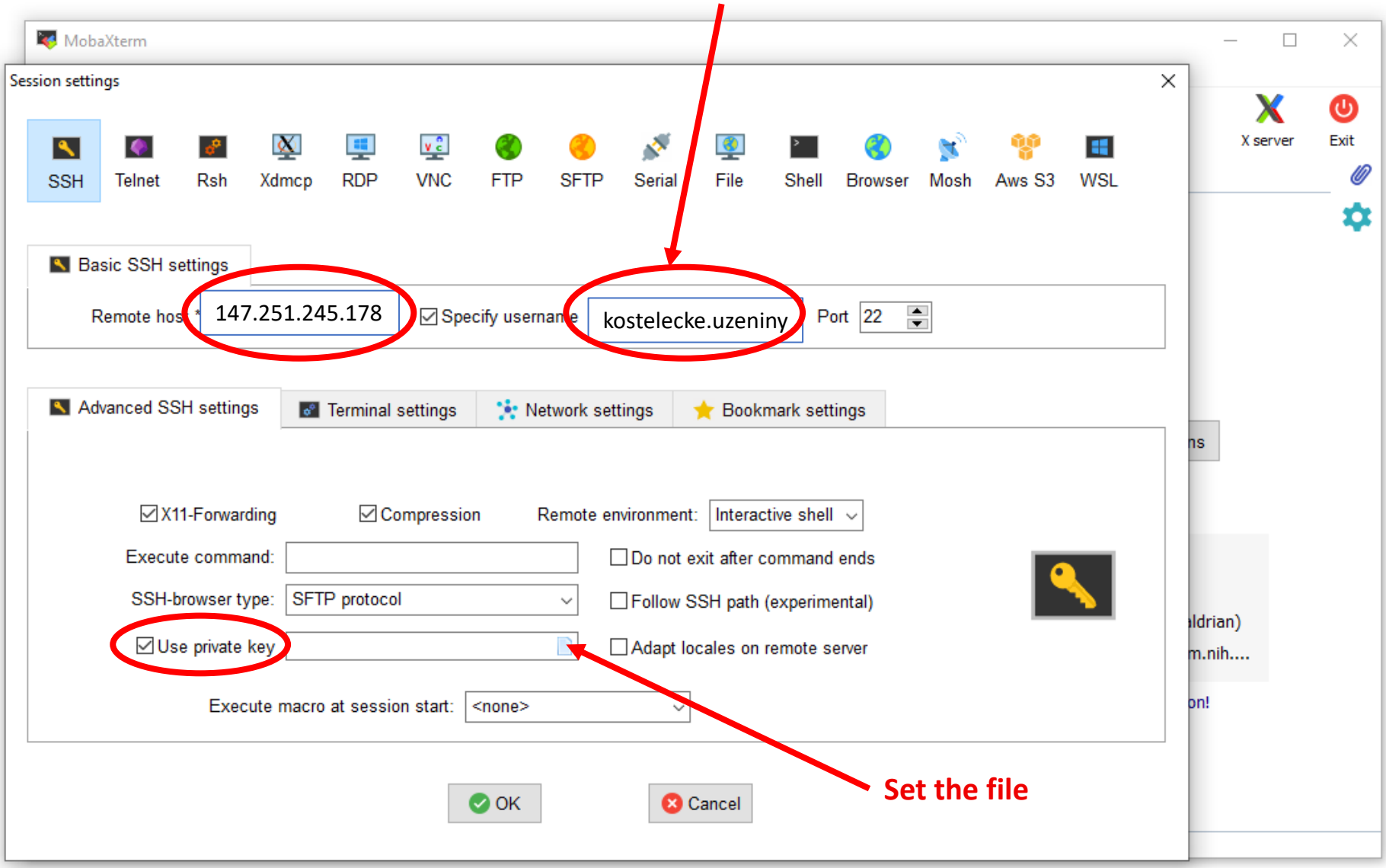


Access to the Linux server



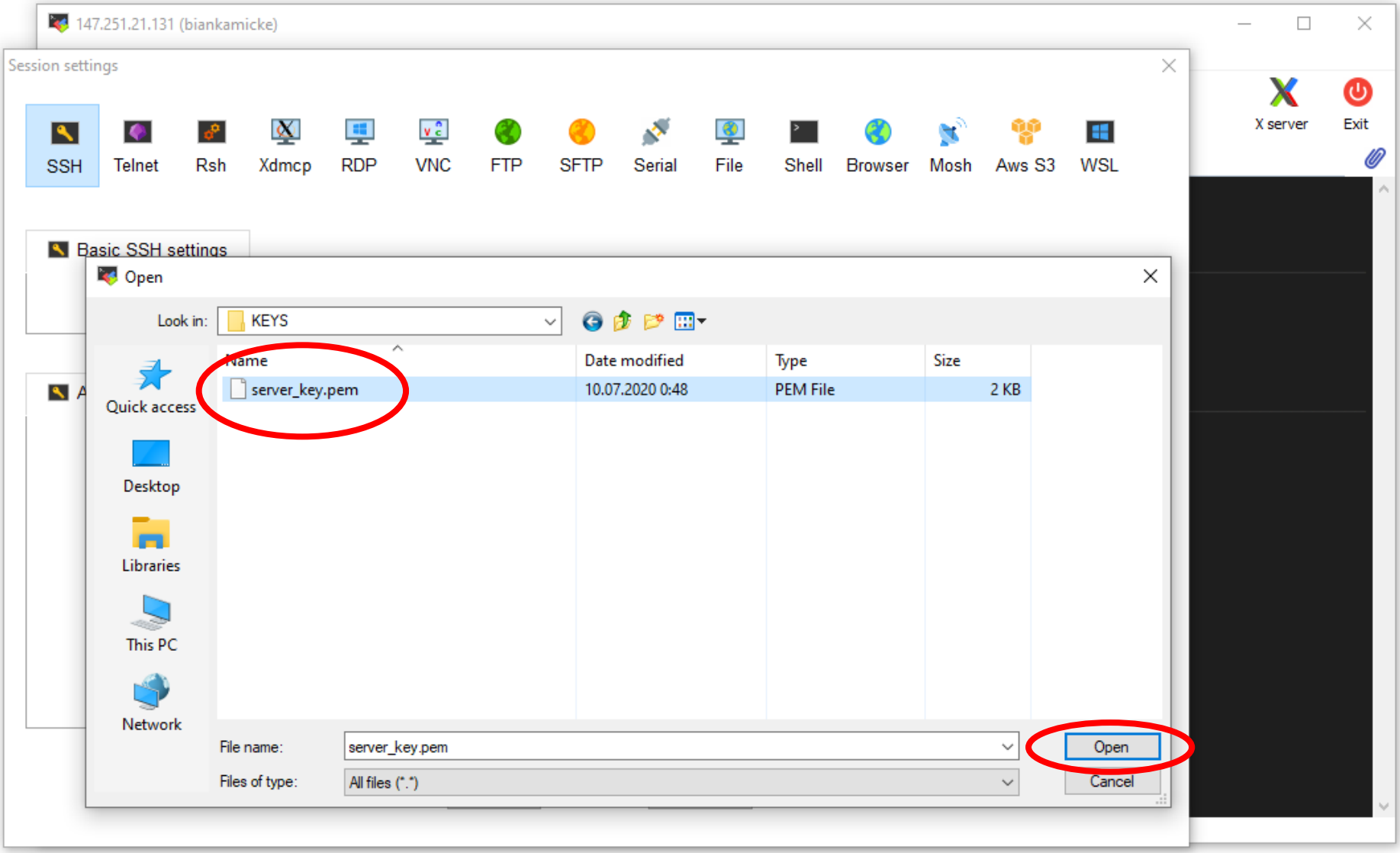
Access to the Linux server

The first part of email before „@“ – my example:
kostelecke.uzeniny@seznam.cz

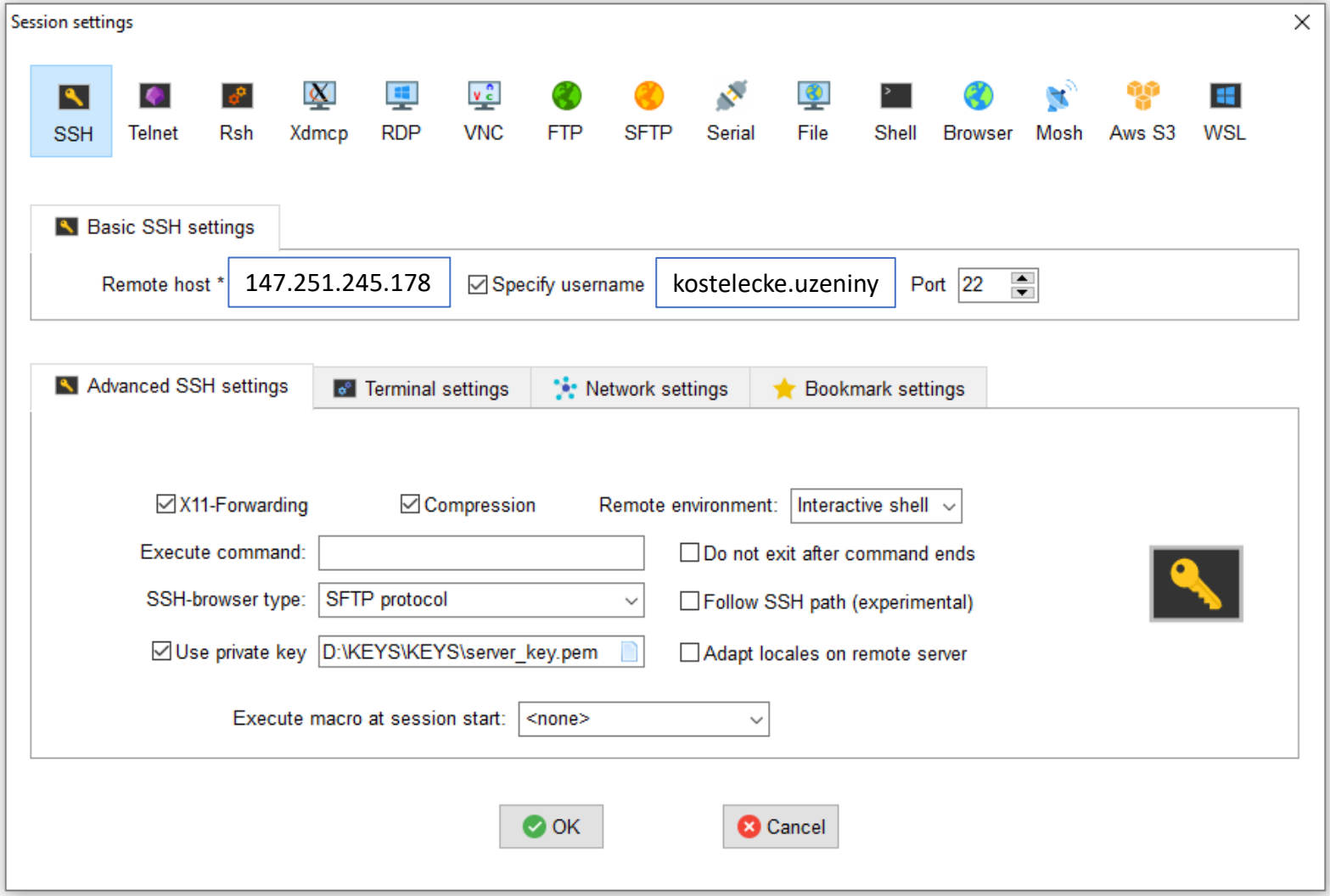


Set the file

Access to the Linux server (key is attached to the email)



Access to the Linux server



Access to the Linux server

6. PROCESSING

```
                    • MobaXterm 20.2 •
            (SSH client, X-server and networking tools)

 ➢ SSH session to [redacted]@147.251.21.131
   • SSH compression : ✔
   • SSH-browser     : ✔
   • X11-forwarding  : ✔  (remote display is forwarded through SSH)
   • DISPLAY         : ✔  (automatically set on remote server)

 ➢ For more info, ctrl+click on help or visit our website

Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.4.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Thu Feb 23 16:26:15 UTC 2023

  System load:  0.0                Processes:             371
  Usage of /:   5.9% of 247.94GB   Users logged in:       0
  Memory usage: 0%                 IPv4 address for ens3: 192.168.0.38
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

53 updates can be applied immediately.
To see these additional updates run: apt list --upgradable


*** System restart required ***
Last login: Thu Feb 23 14:26:57 2023 from [redacted]
[redacted]@intense-computing-machine:~$
```

Access to the Linux server - MAC

ssh -i server_key.pem <user_name>@147.251.245.178

- Open terminal and go the your **.ssh/** directory
- Create a file called **config**
- Add the following 1 line configuration in this file

```
Host * PubkeyAcceptedKeyTypes=+ssh-dss
```

p <path to your key> ~/.ssh/<name of your key>
Now you should be able to modify the permissions normally.

chmod 600 ~/.ssh/<your key's name>
Then ssh using WSL:

ssh -i ~/.ssh/<name of your key> <username>@<ip address>

Basic commands

pwd	Show current address	pwd
man	Show command manual	man chosen_command
cd	Change directory	cd directory_path
ls	List files and directories	-a or –all List hidden files -l or –list show details -h Human readable
mkdir	Create a directory	mkdir folder_Name

$ pwd
/home/<username>

Create a directory called “carnaval”. Check what is inside your new directory.

Copying, moving and deleting

cp	Copy file or folder	cp original_file copy_file
mv	Move/rename file or folder	mv file directory
rm	Remove file or directory	rm file1 rm –r directory
rmdir	Remove directory	rmdir dir1
touch	Create empty file	touch file.txt

Create three new files empty file called “rain.txt”, “people.txt” and “samba.txt”

Now move rain, people and samba to the carnaval!

Copying, moving and deleting

cp	Copy file or folder	cp original_file copy_file
mv	Move/rename file or folder	mv file directory
rm	Remove file or directory	rm file1 rm –r directory
rmdir	Remove directory	rmdir dir1
touch	Create empty file	touch file.txt

Check what is inside the carnaval with the command “ls”.

The rain is killing it. Remove the rain from the carnaval.

See how this is fun?

Great. Now, lets stop with this party and get serious.

Change the name of the carnaval directory to work and remove the samba and the people.

Download a file from the internet

wget

wget -O sequences.zip "http://www.biomed.cas.cz/mbu/lbwrf/sequences.zip"

unzip

unzip sequences.zip

File exibition

cat	Exibit and contatenate	cat file.txt
less	Read a file	less file.txt
more	Read a file	more file.txt
wc	Count the number of lines, characters and bytes of a file	wc -l file.txt [lines of the file] ls | wc -l [how many files are there]
head	First 10 lines of the file	head -n 21 file.txt
tail	Last 10 lines of the file	tail -n 15 file.txt
sort	Order the lines of the file by the user's definition.	sort names > names.sorted

Check the "sequences" with the command less.

Use the key arrows in your keyboard to navigate through this file and the key "q" to exit the less command.

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Count the lines the fasta file.
Take a look in the first 16 lines of fasta file.
Take a look in the last 4 lines of fasta file.

File manipulation

>	Foward standar output	program.pl > report.txt
<	Modify the standar input to a file	program.pl < gene.fasta
|	Allow the combination of commands	ls | wc -l
>>	Adds the output to the end of the file.	programa.pl >> relatorios

Now we are going to generate a sample of our data_set. Take the 16 first lines of the file ”sequences.fas” and forward to a new file called ”small.fasta” in your work directory. If the file doesn’t exist, the forward sign will create it automatically.

Add the last 12 lines from the file ”sequences.fas” to your ”small.fasta”

Use the pipe sigh “|” to combine the command ls and wc and check how many files are inside the work directory

Text manipulation

cat	Concatenate and exibit	cat text1 text2 > text1text2
grep	Search the file line by line for defined expressions	grep ">" genes.fasta
uniq	Remove duplicated lines	sort alfa | uniq -c
cut	Cut input files. Ideal for tables.	cut -d " " f1 alfa
awk	Programming language with resources for text manipulation	awk –F '{print $2 $1}' table.csv
sed	Used to manipulate and transform text.	sed 's/t/u/g' dna.seq > rna.seq

Lets inspect our sequences.
Use grep to search for the a pattern of your interest in your new "fasta" file. For example something from their header. "SampleID_8007"
Count how many times the motive appear in the header by "wc" command.
Choose a pattern, for example the start of the read name ">" and use grep to count how many times it show up in your

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Now to finish, lets see how many commands we learned, using the command history and forwarding the output to the file commands.txt

To inspect this file, use the commands:
cut -f 7 -d ' ' commands.txt | sort | uniq | wc -l
What does it show you?

What if you want to count how many times you used each one?